

Agents & Math Review

EECS 492

Introduction to Artificial
Intelligence

Reminders

- Today we are discussing agents and intro to ML

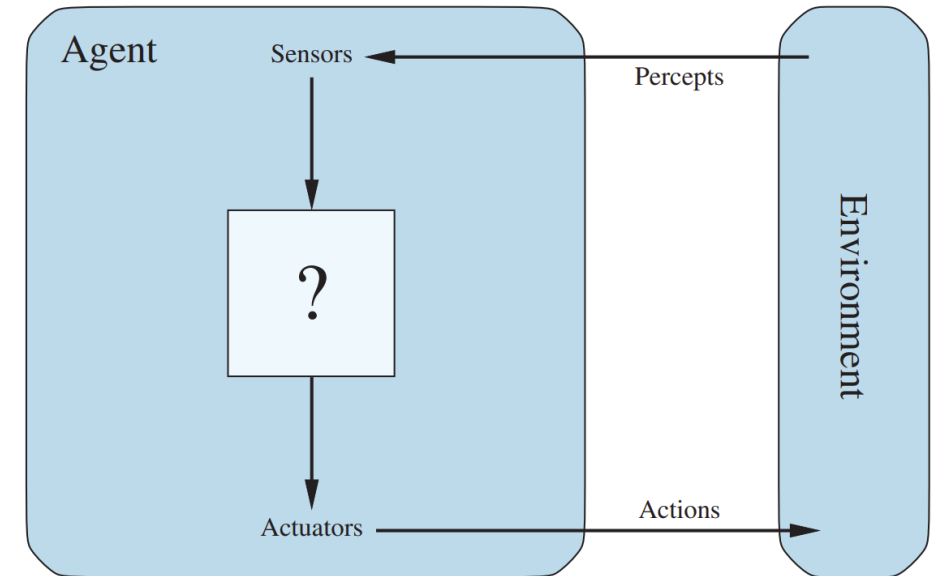
Today's Class

- Agents
- Math review
 - Vectors and Matrices
 - Multivariate calculus
 - Probability

Agents

What is an Agent?

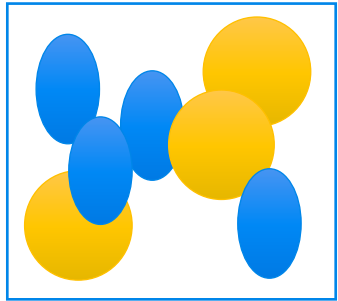
- An agent is anything that:
 - **Perceives** its environment through sensors
 - The things perceived are called *percepts*.
 - **Decides how to act** on its environment
 - Acts via *actuators*.
- An agent pursues a set of pre-specified goals, or a pre-specified strategy
 - Entrusted to act autonomously



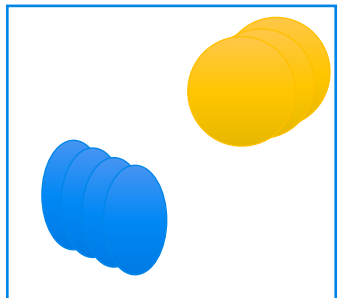
Rational Agents

- The goal is to compute the “right thing to do”
- Maximize **expected performance measure...**
 - How we want the world to be
(not agent actions we think we want)
- ... by considering:
 - Percept sequence
 - Prior knowledge
 - Available actions
- To design an agent,
we first need to understand its environment...

Goal state (desired organization)
not
Goal action (organize the shapes)
(e.g., might repeatedly organize, mess up, organize, mess up, ...)



Start State



Goal State

Describing the Task Environment

- PEAS Description



Performance Measure	Environment	Actuators	Sensors
what somebody wants achieved thanks to the agent's actions	what features of the world are relevant, and what values they can take	how can the agent affect the environment through chosen actions	how can the agent know features of the environment, and how well

Discuss in small groups: Autonomous Shuttle



<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRLqgVZQonVa7XHA5A04TlhwQ-GCLB3XTLTHQ&usqp=CAU>

Performance Measure	Environment	Actuators	Sensors
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Autonomous Shuttle



<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRLqgVZQonVa7XHA5A04TlhwQ-GCLB3XTLTHQ&usqp=CAU>

Performance Measure	Environment	Actuators	Sensors
Safe, fast, legal, comfortable trip, maximize profits, minimize negative impact on others	Roads, other traffic, pedestrians, customers, weather, ...	Steering, accelerator, brake, signal, horn, display, speech, ...	Cameras, radar, speedometer, GPS, engine sensors, accelerometer...

Task Environment

Simplified representations of reality to capture the most important features for solving the problem at hand with AI



Task Environment Dimensions

- Fully observable vs. partially observable
- Deterministic vs. nondeterministic
- Episodic vs. sequential
- Static vs. dynamic
- Discrete vs. continuous
- Known vs. unknown
- Single-agent vs. multi-agent

Task Environment Dimensions

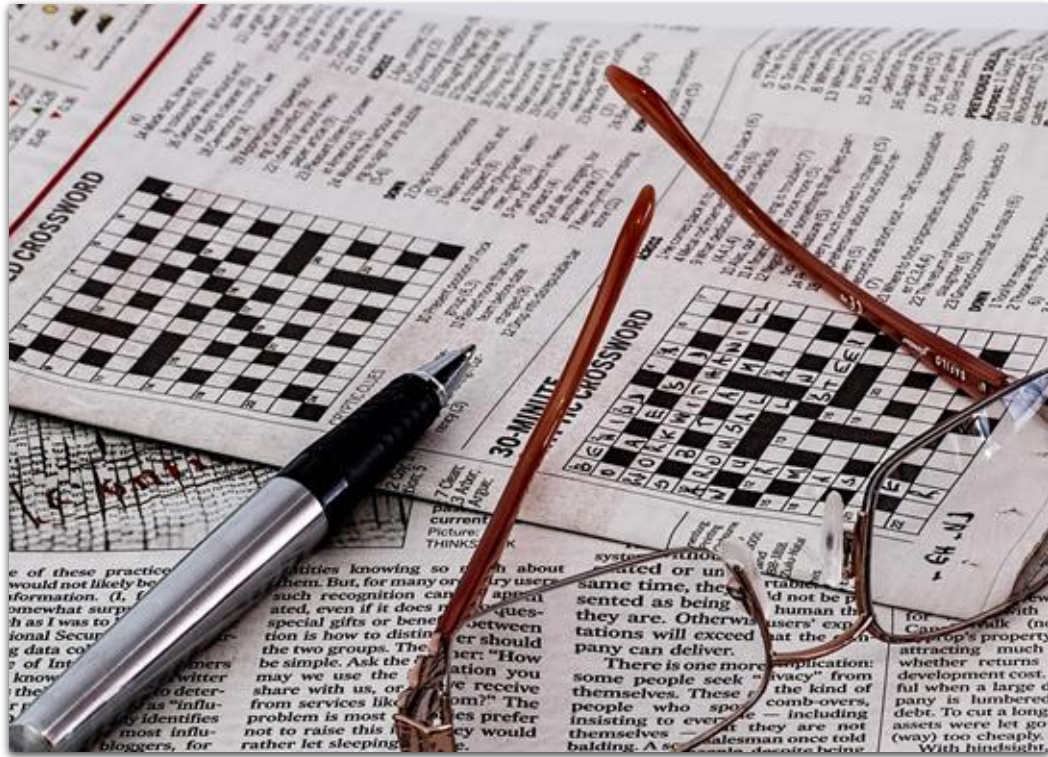
- **Fully observable vs. partially observable**
- Deterministic vs. nondeterministic
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Fully Observable

Partially Observable

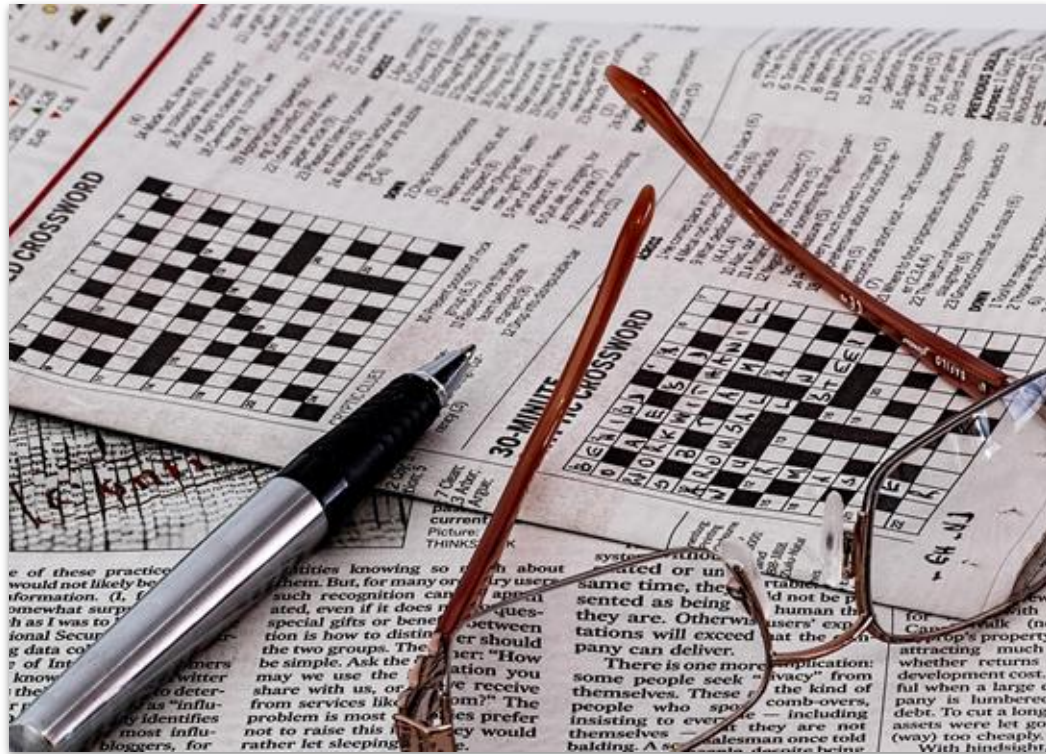


Fully Observable



Partially Observable

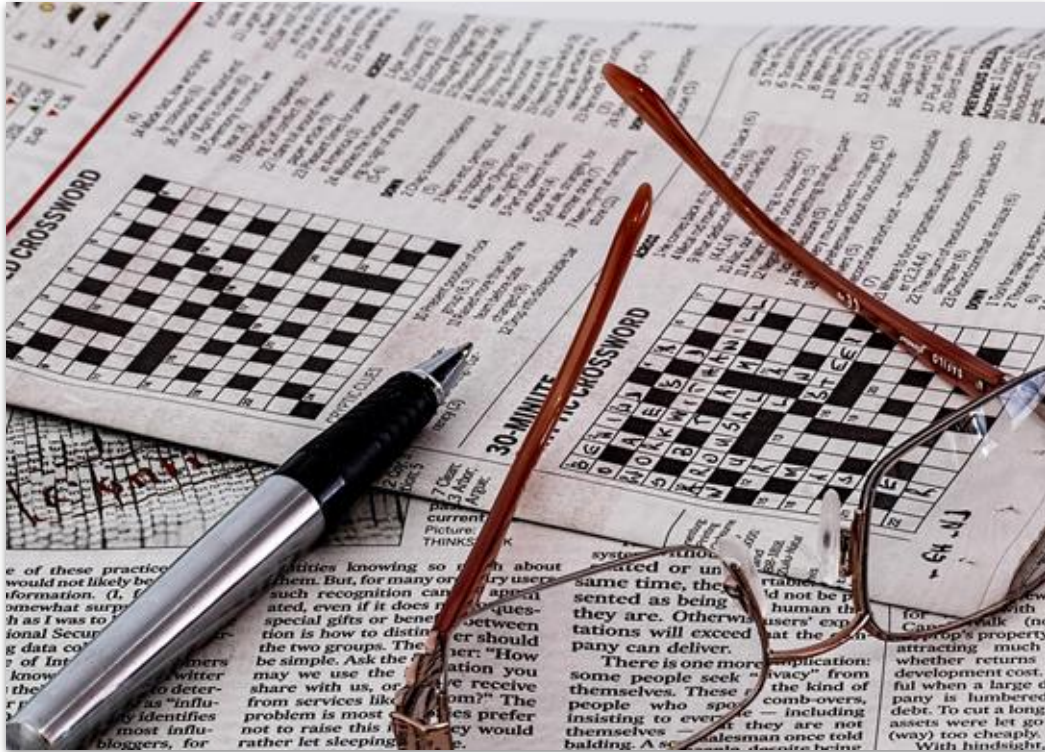
Fully Observable



Partially Observable

There are noisy or inaccurate sensors, or missing information about the state

Fully Observable



Partially Observable



Task Environment Dimensions

- Fully observable vs. partially observable
- **Deterministic vs. nondeterministic**
- Episodic vs. sequential
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Deterministic

Next state of the environment is
completely determined by current
state and action

Nondeterministic

Deterministic

Nondeterministic



Deterministic



Nondeterministic

Next state of the environment is
uncertain or unobserved

Deterministic



Nondeterministic



Task Environment Dimensions

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Episodic

Sequential

An agent's experience is divided into
atomic **episodes**



- Receives a percept
- Performs a single action
- Next episode doesn't depend on actions in the previous one

Episodic

Sequential



Episodic



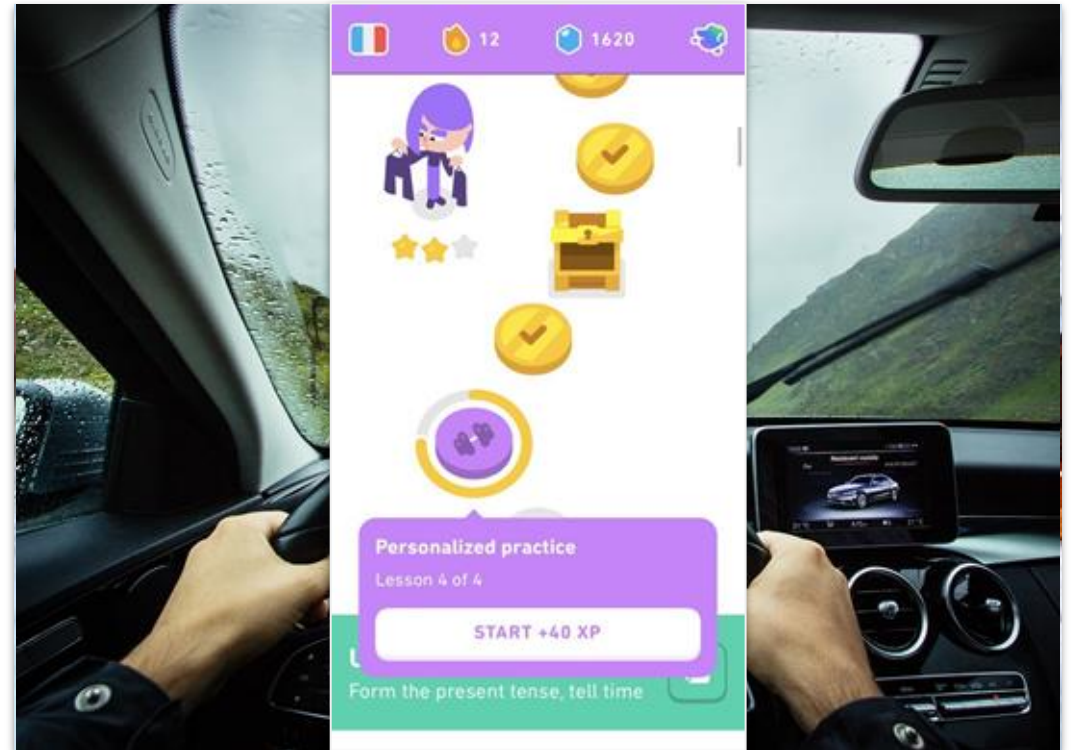
Sequential

Current decision could affect all
future decisions

Episodic



Sequential



Task Environment Dimensions

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- Episodic vs. sequential
- **Static vs. dynamic**
- Discrete vs. continuous
- Known vs. unknown
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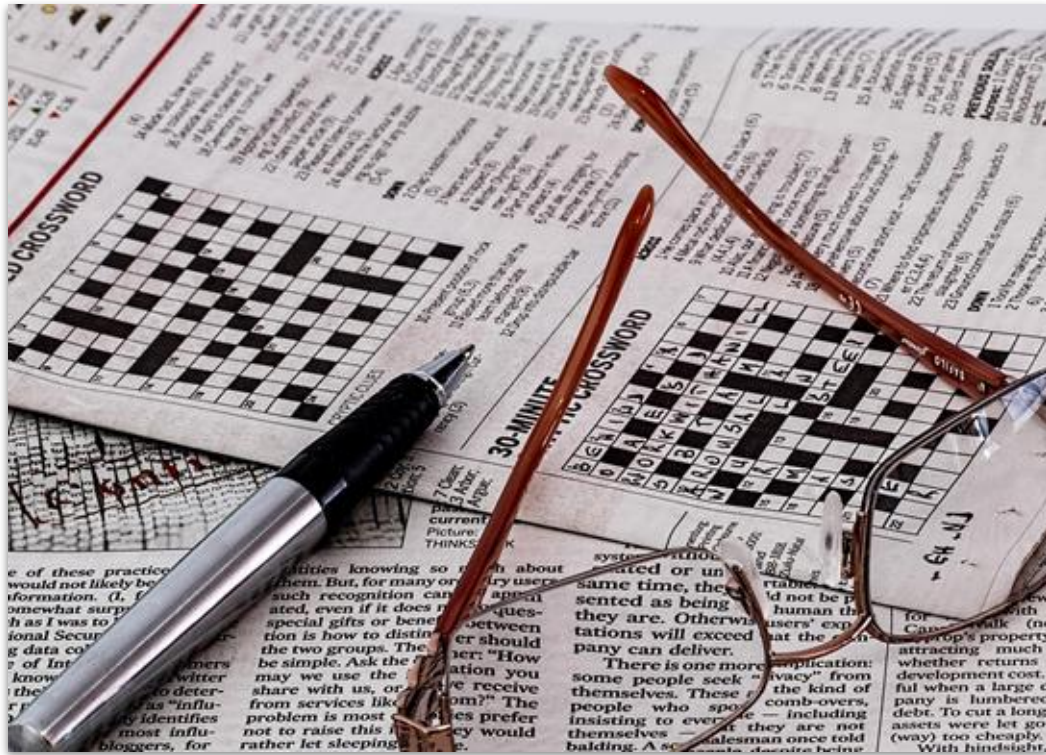
Static

World isn't changing while agent
decides on action

Dynamic

Static

Dynamic



Static



Dynamic

World is changing while agent is
deciding on action

Static



Dynamic



Task Environment Dimensions

- Fully observable vs. partially observable
- Deterministic vs. nondeterministic
- Episodic vs. sequential
- Static vs. dynamic
- **Discrete vs. continuous**
- Known vs. unknown
- Single-agent vs. multi-agent

Discrete

Continuous

Finite number of states, percepts,
and/or actions

Discrete

Continuous



Discrete



Continuous

Infinite number of states, percepts,
and/or actions

Discrete



Continuous

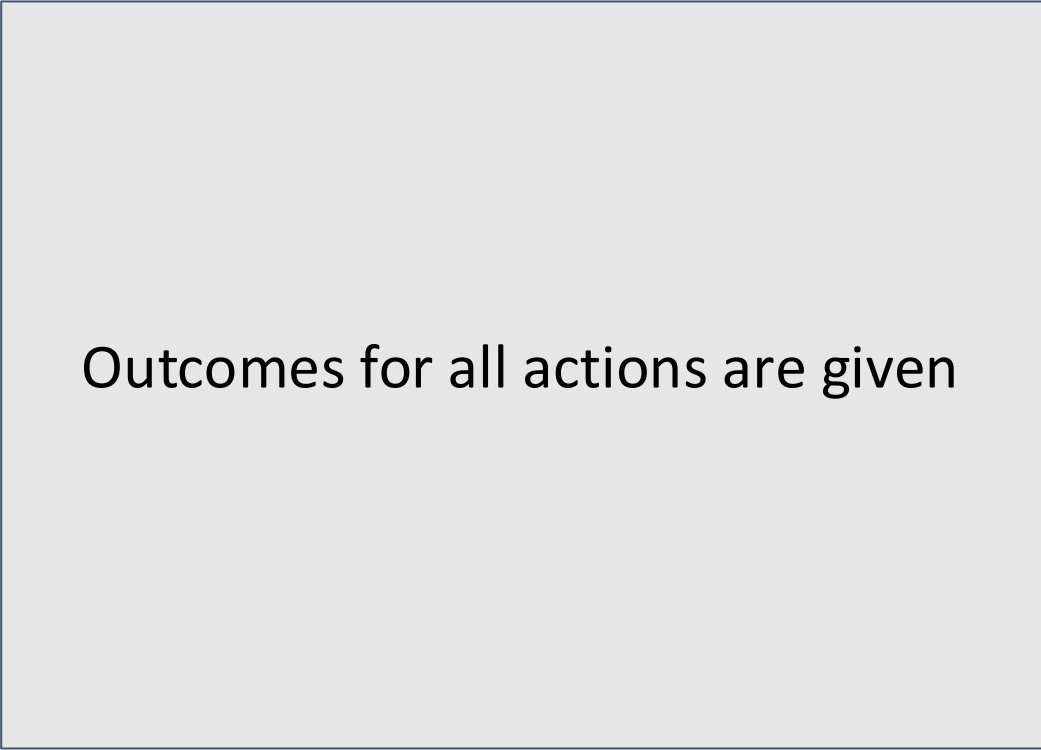


Task Environment Dimensions

- Fully observable vs. partially observable
- Deterministic vs. nondeterministic
- Episodic vs. sequential
- Static vs. dynamic
- Discrete vs. continuous
- **Known vs. unknown**
- Single-agent vs. multi-agent

Known

Unknown



Outcomes for all actions are given

Known



Unknown

Known



Unknown

Agent needs to learn how
environment works

Known



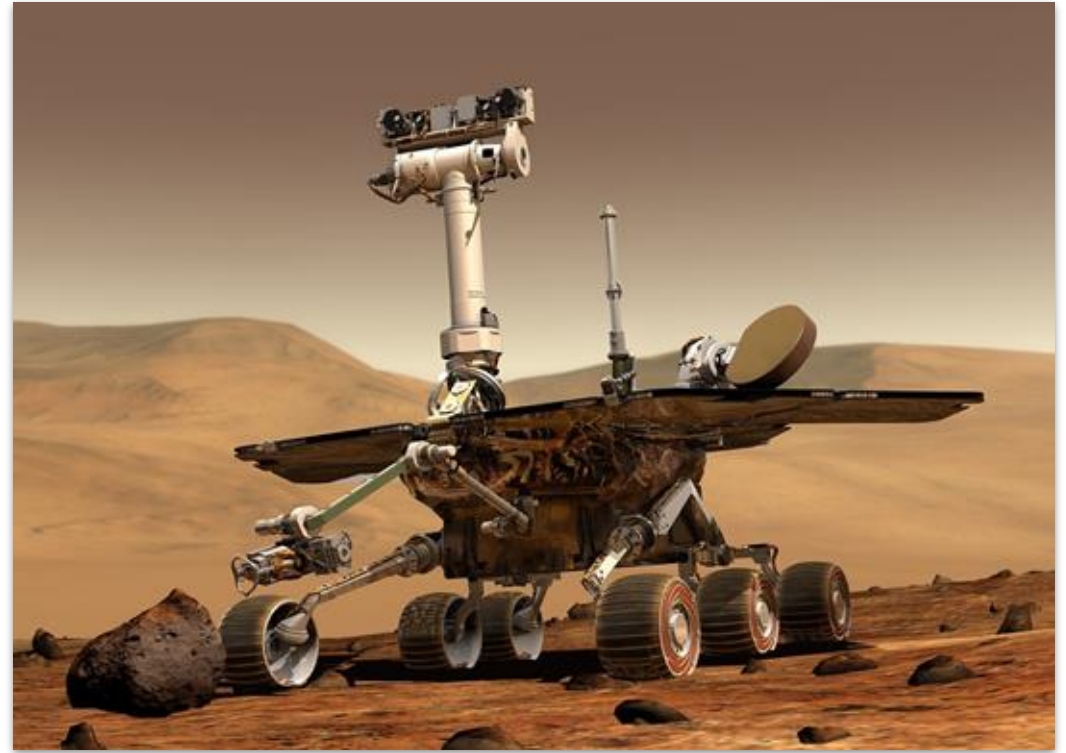
Unknown



Known



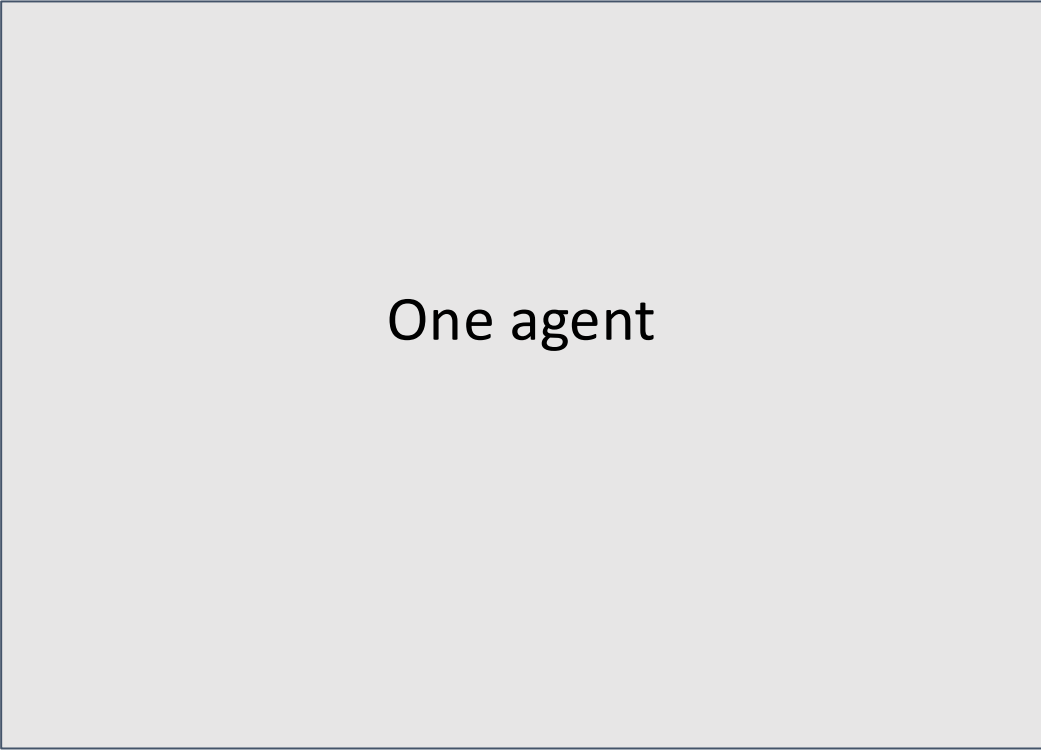
Unknown



Task Environment Dimensions

- Fully observable vs. partially observable
- Deterministic vs. nondeterministic
- Episodic vs. sequential
- Static vs. dynamic
- Discrete vs. continuous
- Known vs. unknown
- **Single-agent vs. multi-agent**

Single-agent



One agent

Multi-agent

Single-agent

Multi-agent



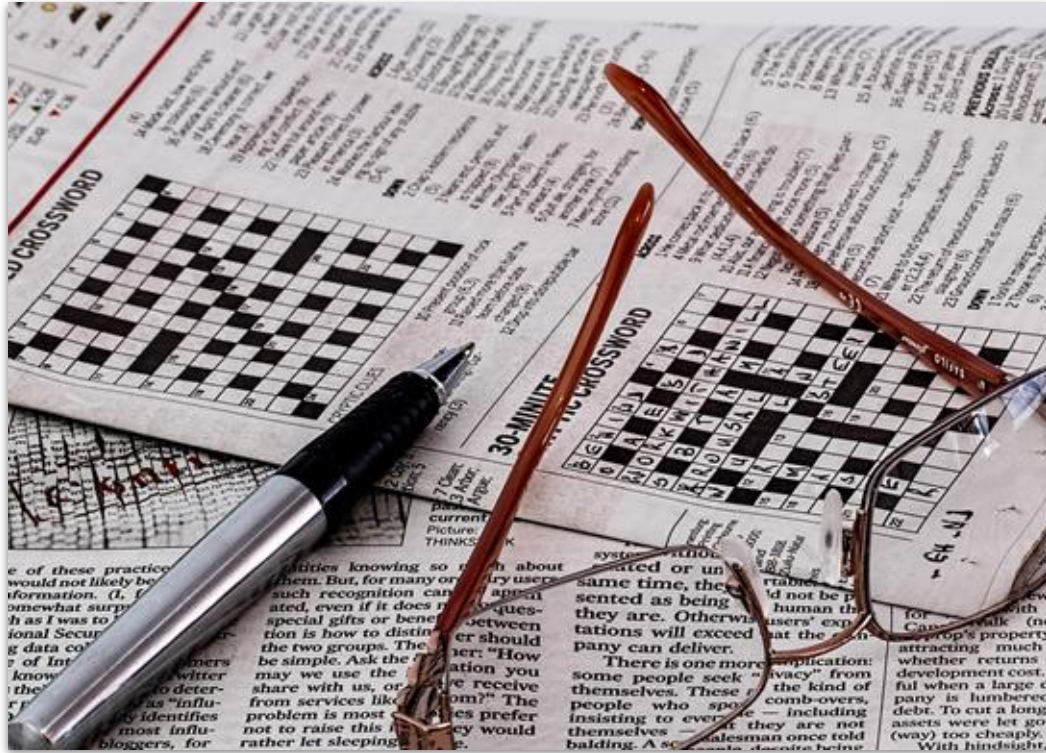
Single-agent



Multi-agent

More than one agent

Single-agent



Multi-agent



Single-agent



Multi-agent



Hard Task Environments

- Fully observable vs. **partially observable**
- Deterministic vs. **nondeterministic**
- Episodic vs. **sequential**
- Static vs. **dynamic**
- Discrete vs. **continuous**
- Known vs. **unknown**
- Single-agent vs. **multi-agent**

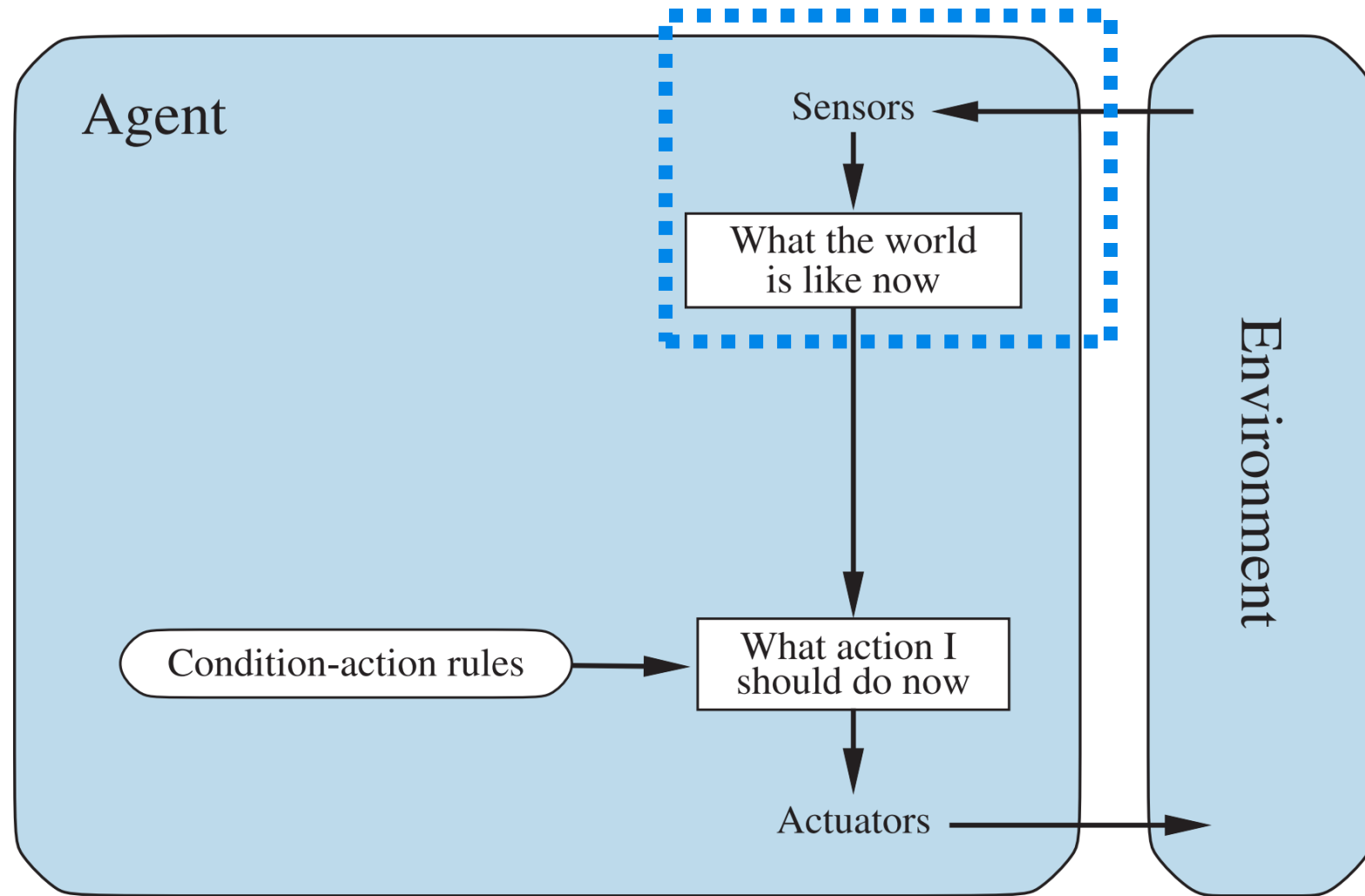
Ambiguity



- Episodic vs. sequential?
- Single-agent vs. multi-agent?

Simple Reflex Agent

React based on
current percept only.

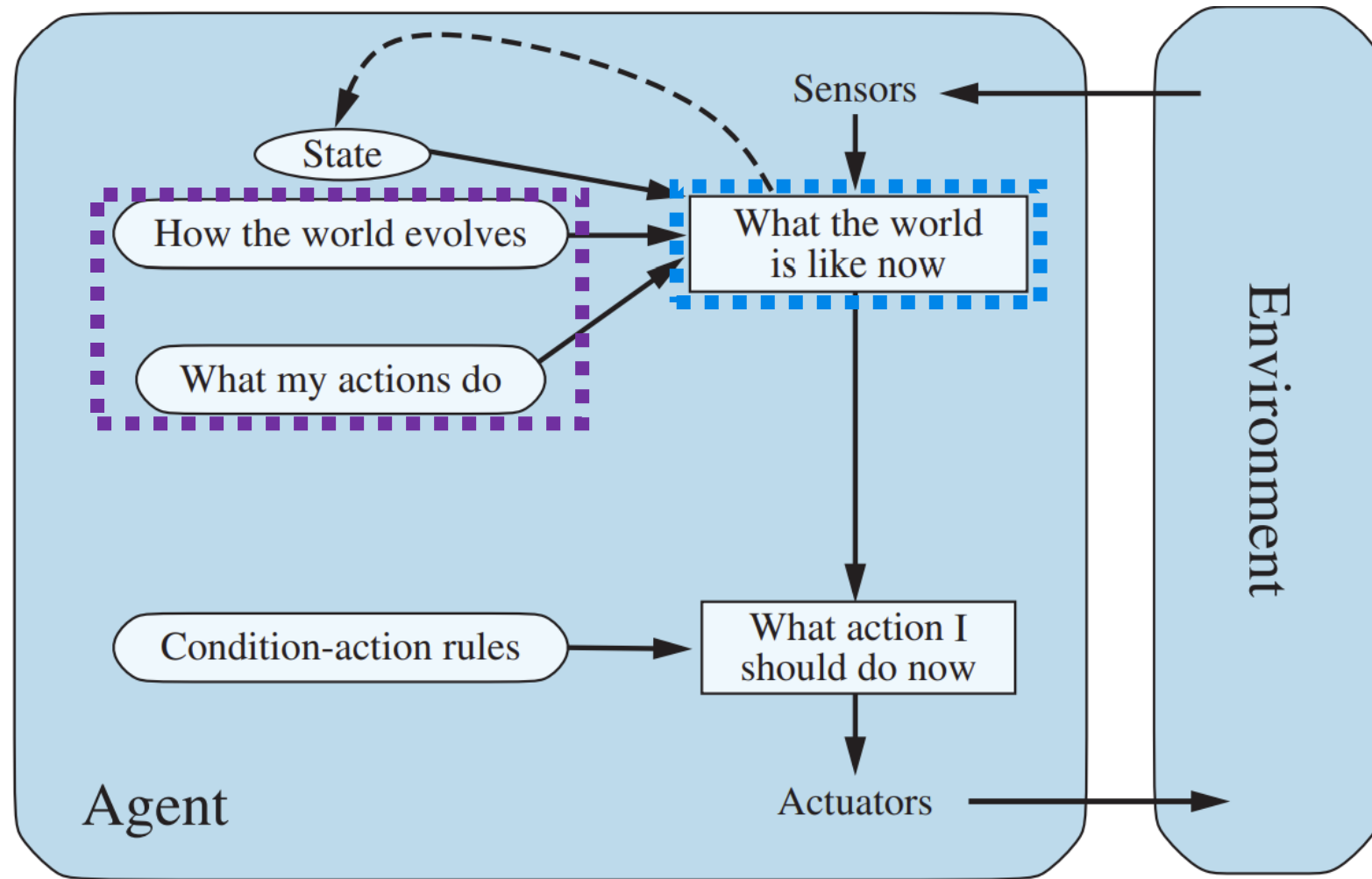


If **floor is dirty below me**, clean there.
else if ...

Model-Based Reflex Agent

Also:

- Info about **how the world changes**.
- Maintain **internal belief** of the state of the world right now.

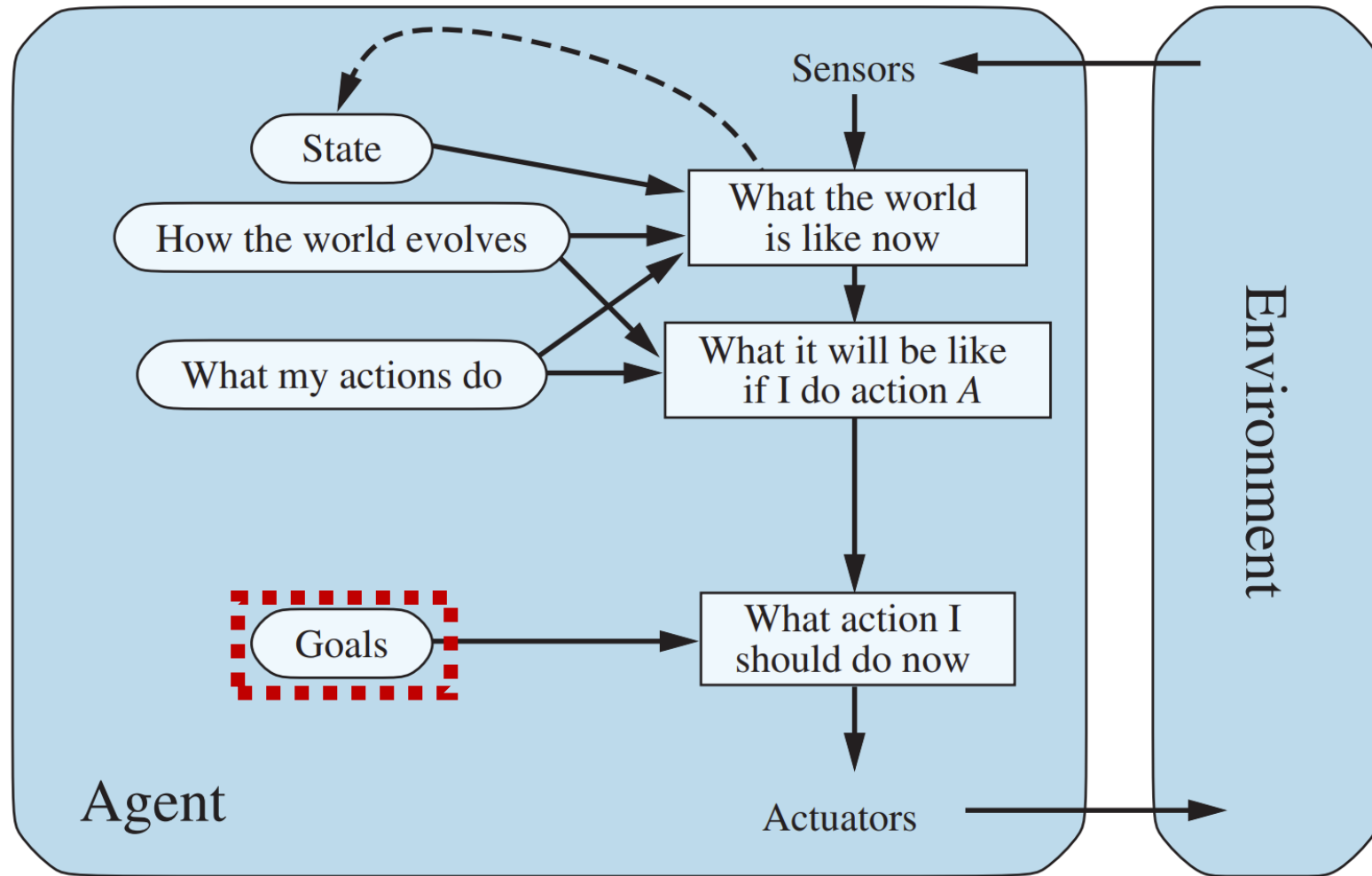


I just cleaned in a spot, so the floor is no longer dirty there.

Model- & Goal-Based Agent

Also:

- Info about **desirable world states**.

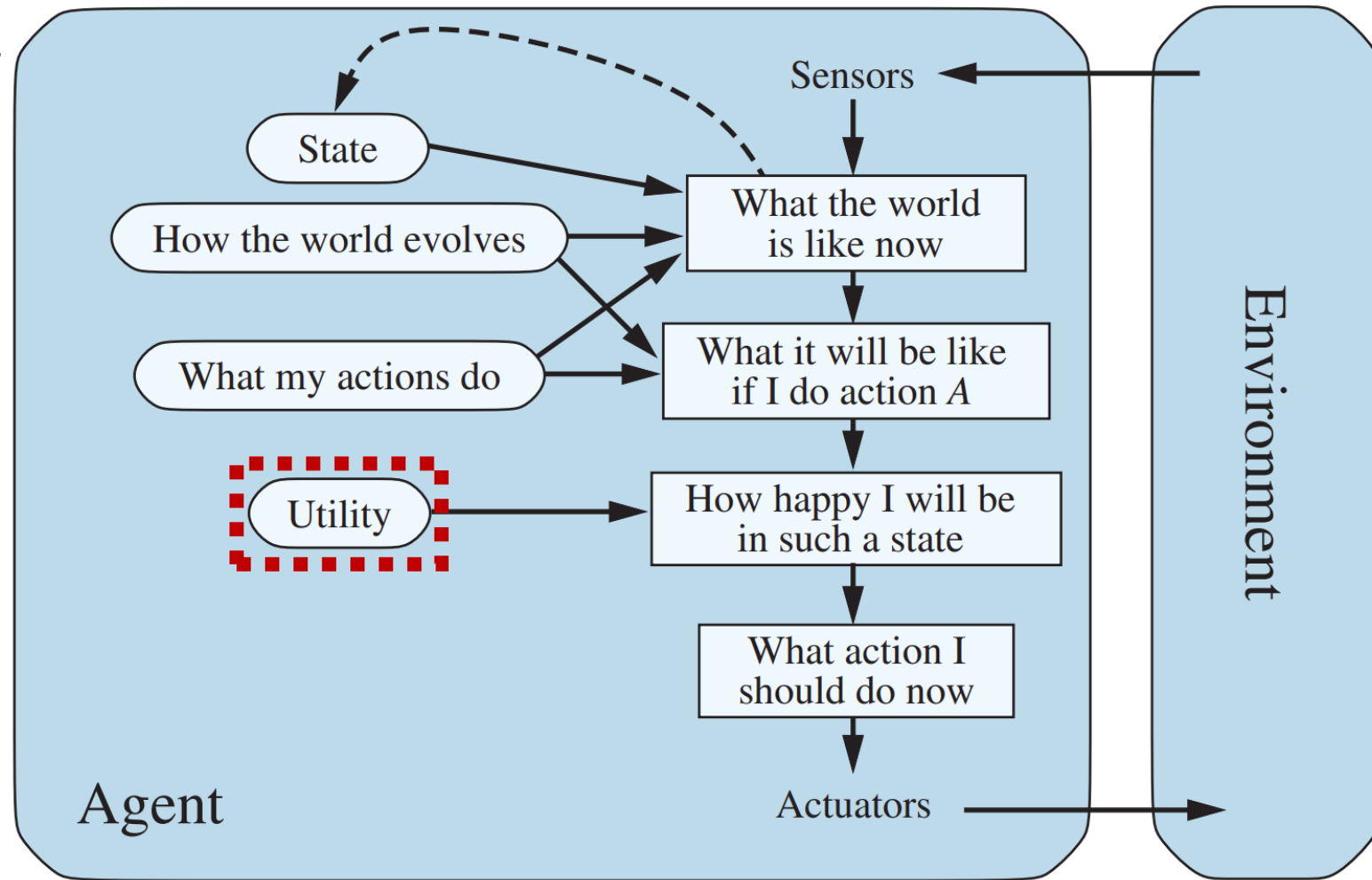


The most desirable state
is one in which the entire floor is clean.

Model- & Utility-Based Agent

Also:

- **How desirable a *path*** to a goal state is.

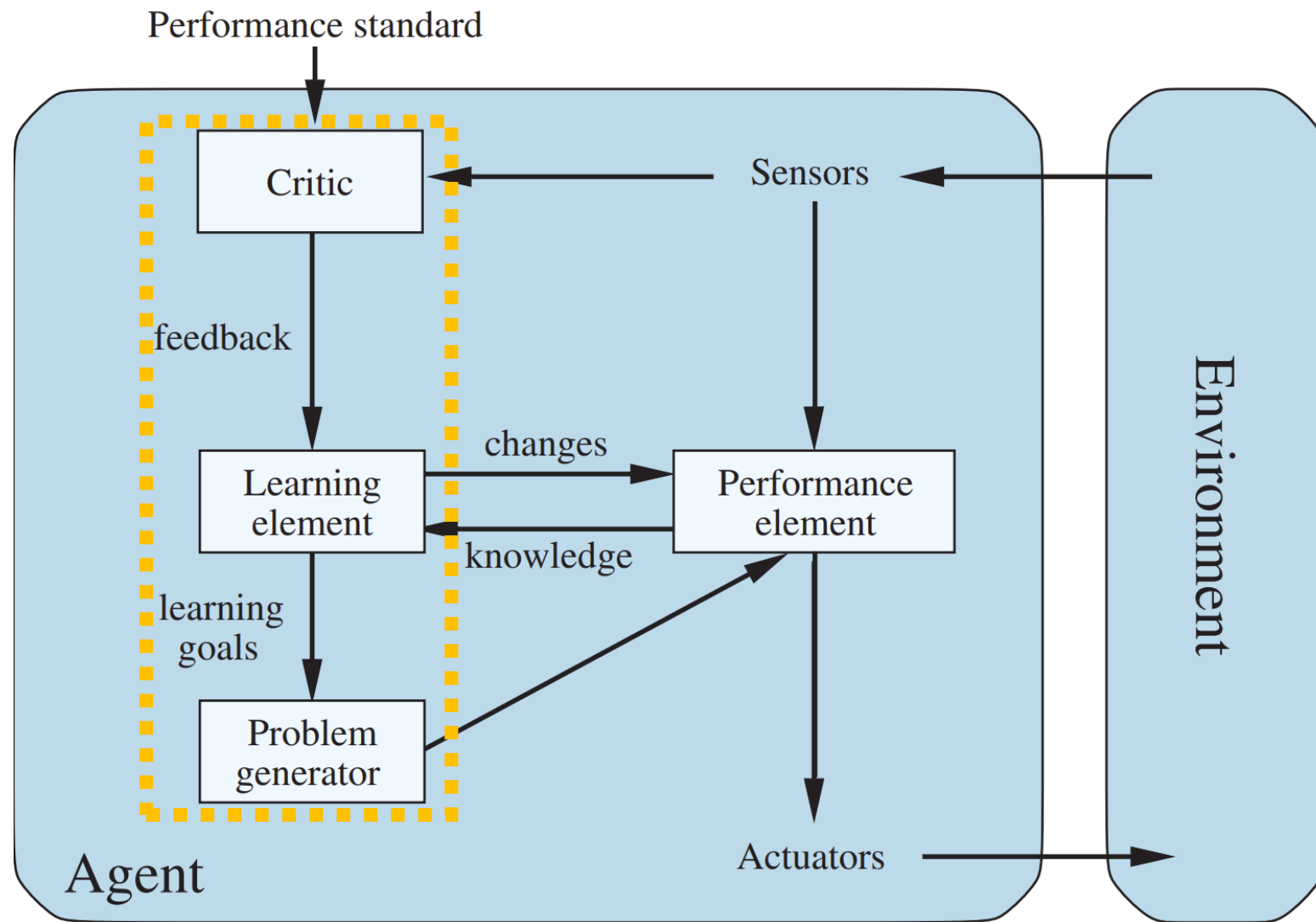


My work will be more useful
if I don't get anywhere near human feet on my way to the goal state.

Learning Agent

Also:

- **Learn** about environment over time.



Don't clean mid-afternoon [*baby nap time*].
Under dining table, especially on one side [*baby's seat*] is extra dirty.

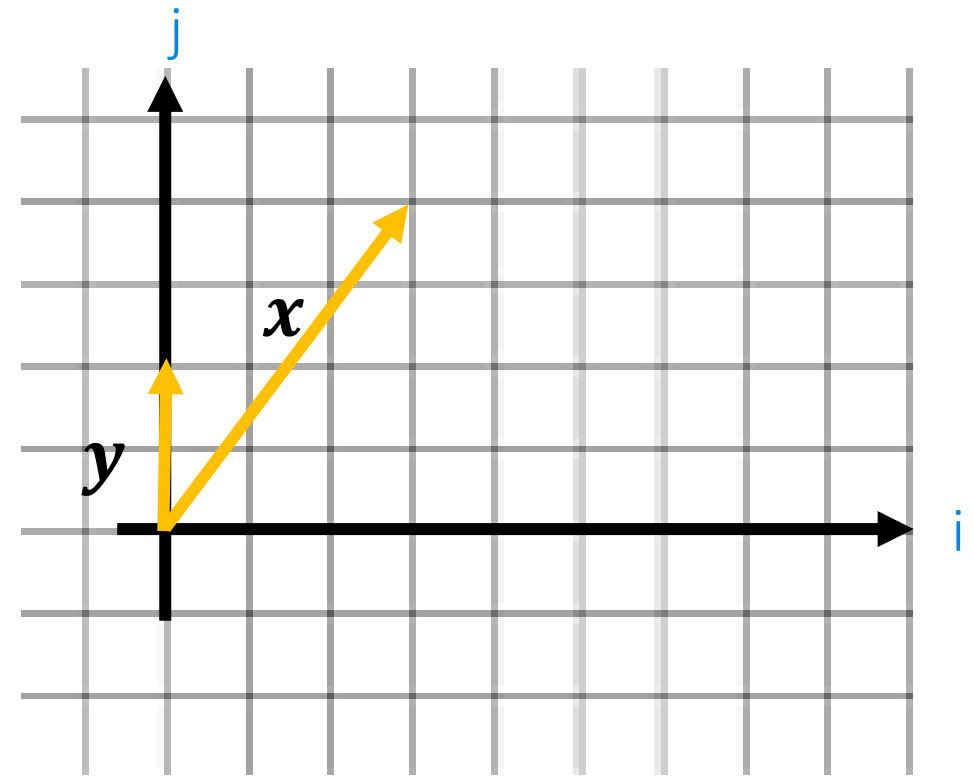
Break time

03:00

Review of Vectors and Matrices

Vector Basics

- **Vector** – ordered sequence of values
 - $\mathbf{x} = \langle 3, 4 \rangle$
 - $\mathbf{y} = \langle 0, 2 \rangle$
 - We will put vector names in **bold**
 - Elements can be accessed using subscripts
 - $\mathbf{z} = \langle z_1, z_2, \dots, z_n \rangle$
- Vector addition – elementwise sum
 - $\mathbf{x} + \mathbf{y} = \langle 3 + 0, 4 + 2 \rangle = \langle 3, 6 \rangle$
- Scalar multiplication – multiplies each element by a constant
 - $5\mathbf{x} = \langle 5 \times 3, 5 \times 4 \rangle = \langle 15, 20 \rangle$



Notation: Vectors

- We can consider a *row vector*: $\mathbf{a} = [a_1 \quad a_2 \quad \cdots \quad a_n]$
- But the default view of a vector is a *column vector*:

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = [b_1 \quad b_2 \quad \cdots \quad b_n]^T$$

Sometimes, people write it this way to conveniently fit on one line.
 T means ***transpose***.

Notation: Vectors

Often, a **subscript** is used to indicate one **component** of the vector.

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

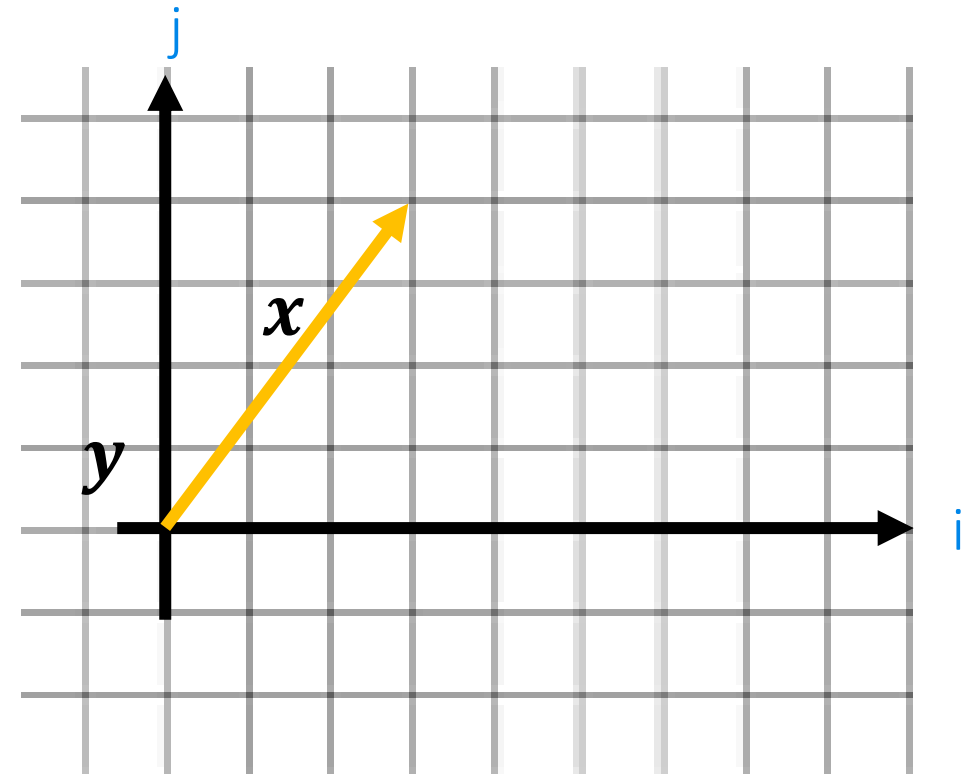
Multiple vectors with the same variable name but each with a different "number" are differentiated via **parenthesized superscript (i)**:

$$\mathbf{b}^{(1)} = \begin{bmatrix} b_1^{(1)} \\ b_2^{(1)} \\ \vdots \\ b_n^{(1)} \end{bmatrix} \quad \mathbf{b}^{(2)} = \begin{bmatrix} b_1^{(2)} \\ b_2^{(2)} \\ \vdots \\ b_n^{(2)} \end{bmatrix} \quad \mathbf{b}^{(3)} = \begin{bmatrix} b_1^{(3)} \\ b_2^{(3)} \\ \vdots \\ b_n^{(3)} \end{bmatrix}$$

This does **not** indicate exponentiation!

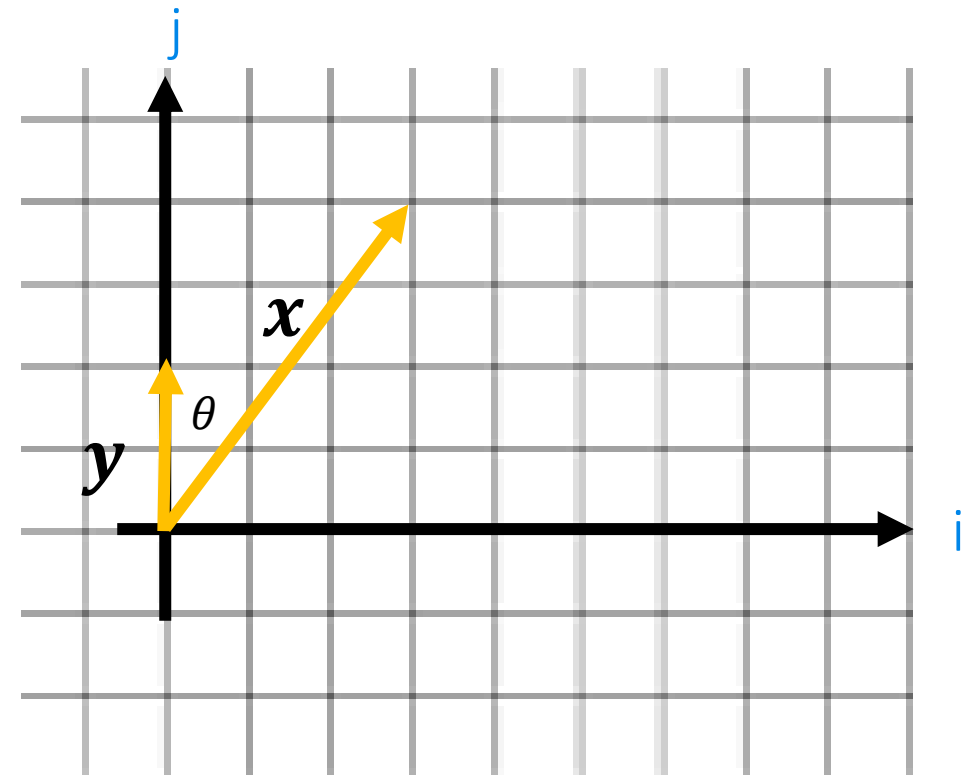
Vector Length

- Vector length – denoted $|\mathbf{x}|$
 - $\mathbf{x} = \langle 3, 4 \rangle$
 - $|\mathbf{x}| = \sqrt{3^2 + 4^2} = 5$



Vector Dot Product

- Dot product (or scalar product) $\mathbf{x} \cdot \mathbf{y}$
 - $\mathbf{x} = \langle 3, 4 \rangle$
 - $\mathbf{y} = \langle 0, 2 \rangle$
 - $\mathbf{x} \cdot \mathbf{y} = \sum_i x_i y_i = 3 \times 0 + 4 \times 2 = 8$
- Also:
 - $\mathbf{x} \cdot \mathbf{y} = |\mathbf{x}| |\mathbf{y}| \cos \theta$
 - θ is the angle between $\mathbf{x}$ and $\mathbf{y}$



Matrices

- **Matrix** – rectangular array of values arranged in rows & columns

- Matrix $\mathbf{X}$ of size $n_x \times m$:

$$\mathbf{X} = \begin{bmatrix} x_1^{(1)} & x_1^{(2)} & \cdots & x_1^{(m)} \\ x_2^{(1)} & x_2^{(2)} & \cdots & x_2^{(m)} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n_x}^{(1)} & x_{n_x}^{(2)} & \cdots & x_{n_x}^{(m)} \end{bmatrix}$$

- First index specifies row, second index specifies column

From Vectors to Matrices

Suppose we have m vectors,
each of length n_x .

At times, it will be convenient to put
them together into a matrix.

Matrix variables are traditionally
bold-faced capital letters.

n_x rows
 m columns
 $\mathbf{X}$ has *shape* (n_x, m)
 $\mathbf{X} \in \mathbb{R}^{n_x \times m}$

$$\begin{aligned}
 \mathbf{x}^{(1)} &= \begin{bmatrix} x_1^{(1)} \\ x_2^{(1)} \\ \vdots \\ x_{n_x}^{(1)} \end{bmatrix} & \mathbf{x}^{(2)} &= \begin{bmatrix} x_1^{(2)} \\ x_2^{(2)} \\ \vdots \\ x_{n_x}^{(2)} \end{bmatrix} & \dots & \mathbf{x}^{(m)} = \begin{bmatrix} x_1^{(m)} \\ x_2^{(m)} \\ \vdots \\ x_{n_x}^{(m)} \end{bmatrix} \\
 & \vdots & & & & \\
 \mathbf{X} &= \begin{bmatrix} x_1^{(1)} & x_1^{(2)} & \dots & x_1^{(m)} \\ x_2^{(1)} & x_2^{(2)} & \dots & x_2^{(m)} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n_x}^{(1)} & x_{n_x}^{(2)} & \dots & x_{n_x}^{(m)} \end{bmatrix} \\
 &= \begin{bmatrix} | & | & \dots & | \\ \mathbf{x}^{(1)} & \mathbf{x}^{(2)} & \dots & \mathbf{x}^{(m)} \\ | & | & \dots & | \end{bmatrix} & \mathbf{x}^{(i)} \in \mathbb{R}^{n_x} & \text{The } | \text{ are informal,} \\
 &= [\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(m)}] & & \text{intuitive notation.}
 \end{aligned}$$

We could call this a matrix of column vectors.

Matrix Operations

- Matrix addition – sum of corresponding elements
 - $(\mathbf{A} + \mathbf{B})_{i,j} = \mathbf{A}_{i,j} + \mathbf{B}_{i,j}$
 - Sum is undefined if $\mathbf{A}$ and $\mathbf{B}$ have different sizes
- Multiplication by a scalar
 - $(c\mathbf{A})_{i,j} = c\mathbf{A}_{i,j}$

Examples with 2x2 matrices

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} a + e & b + f \\ c + g & d + h \end{pmatrix}$$

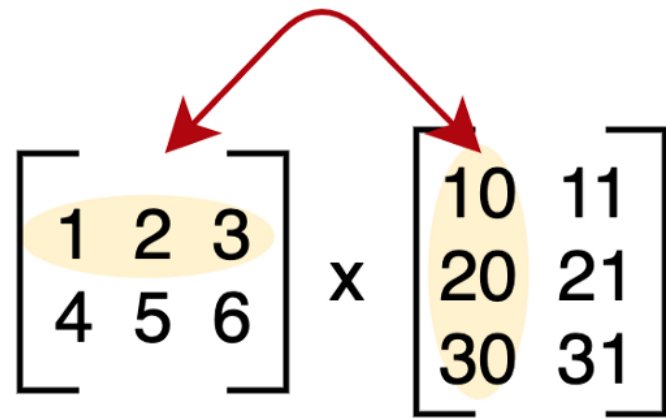
$$k \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} ka & kb \\ kc & kd \end{pmatrix}$$

More on Matrix Multiplication

$$\begin{matrix}
 \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix} & \begin{bmatrix} c_1 & c_2 & \dots & c_p \end{bmatrix} & = & \begin{bmatrix} r_1 \cdot c_1 & r_1 \cdot c_2 & \dots & r_1 \cdot c_p \\ r_2 \cdot c_1 & r_2 \cdot c_2 & \dots & r_2 \cdot c_p \\ \vdots & \vdots & \ddots & \vdots \\ r_m \cdot c_1 & r_m \cdot c_2 & \dots & r_m \cdot c_p \end{bmatrix} \\
 (m, n) & (n, p) & & (m, p) \\
 \uparrow \text{match} \uparrow & & &
 \end{matrix}$$

- Properties of matrix multiplication
 - In general, $\mathbf{AB} \neq \mathbf{BA}$ (not commutative)
 - It is associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$

Matrix Multiplication Example


$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 10 & 11 \\ 20 & 21 \\ 30 & 31 \end{bmatrix}$$
$$= \begin{bmatrix} 1 \times 10 + 2 \times 20 + 3 \times 30 & 1 \times 11 + 2 \times 21 + 3 \times 31 \\ 4 \times 10 + 5 \times 20 + 6 \times 30 & 4 \times 11 + 5 \times 21 + 6 \times 31 \end{bmatrix}$$
$$= \begin{bmatrix} 10+40+90 & 11+42+93 \\ 40+100+180 & 44+105+186 \end{bmatrix} = \begin{bmatrix} 140 & 146 \\ 320 & 335 \end{bmatrix}$$

Identity Matrix

- Identity matrix ***I***
 - $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
 - Identity matrix can be any square matrix – the diagonals are always 1
- For all matrices ***A***, ***AI = A***

Matrix Transpose

- Transpose of $\mathbf{X}$ is $\mathbf{X}^T$
 - Switch rows into columns and vice versa
- Example:

$$\mathbf{A} = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}_{2 \times 3} \quad \mathbf{A}^T = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}_{3 \times 2}$$

Image from <https://www.cuemath.com/algebra/transpose-of-a-matrix/>

$$\mathbf{X}_{(n_x, m)} = \begin{bmatrix} x_1^{(1)} & x_1^{(2)} & \cdots & x_1^{(m)} \\ x_2^{(1)} & x_2^{(2)} & \cdots & x_2^{(m)} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n_x}^{(1)} & x_{n_x}^{(2)} & \cdots & x_{n_x}^{(m)} \end{bmatrix}$$

$$\mathbf{X}^T_{(m, n_x)} = \begin{bmatrix} x_1^{(1)} & x_2^{(1)} & \cdots & x_{n_x}^{(1)} \\ x_1^{(2)} & x_2^{(2)} & \cdots & x_{n_x}^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ x_1^{(m)} & x_2^{(m)} & \cdots & x_{n_x}^{(m)} \end{bmatrix}$$

Matrix Inverse

- For a square matrix $\mathbf{A}$, its inverse is another square matrix $\mathbf{A}^{-1}$
 - $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$
 - Not all square matrices have an inverse
 - If a square matrix has an inverse, it can be computed in $O(n^3)$ time

Review of Multivariate Calculus

Functions

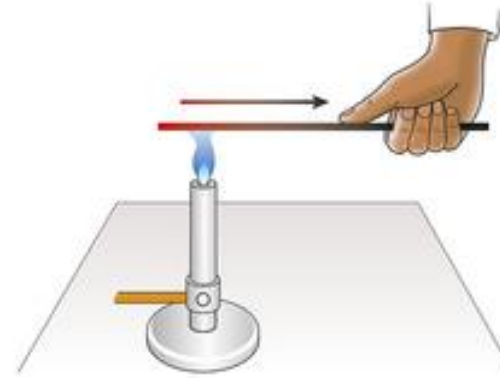
- A **scalar function** maps a single real-valued input to a single real-valued output

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

- **Multivariate scalar function** maps multiple inputs to a single output

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

Temperature as function of x



Elevation as function of x and y



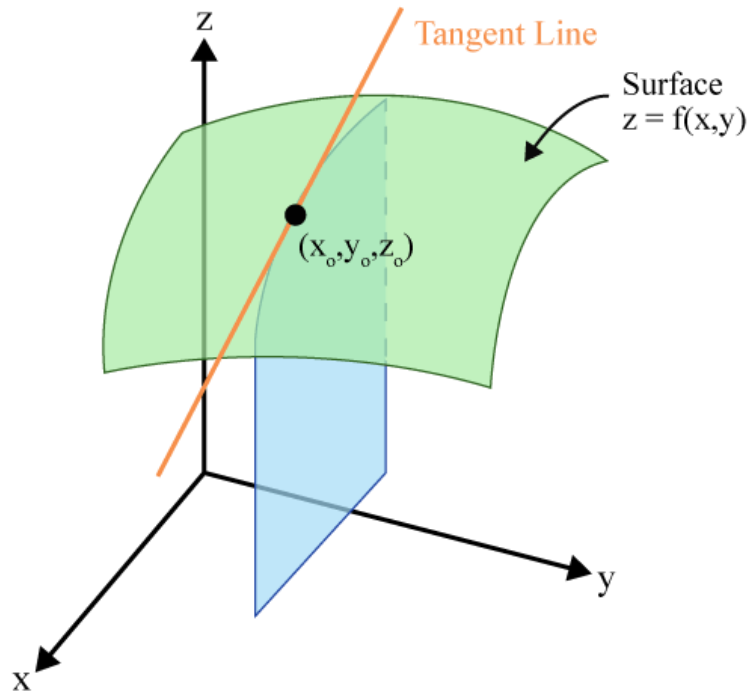
Derivatives of a **scalar** function

- It is the ratio of the change in the output to the change in the input, taken in the limit as the change in the input approaches zero

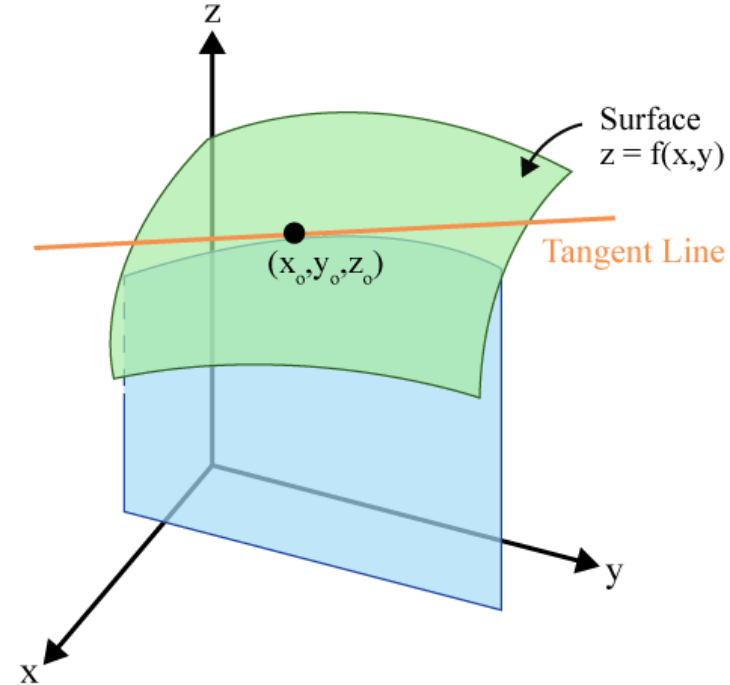
$$f'(x) = \lim_{\varepsilon \rightarrow 0} \frac{f(x + \varepsilon) - f(x)}{\varepsilon}$$

change in output
&
change in input

Partial Derivatives of multivariate scalar function



Slope of the surface in the x -direction



Slope of the surface in the y -direction

Calcworkshop.com

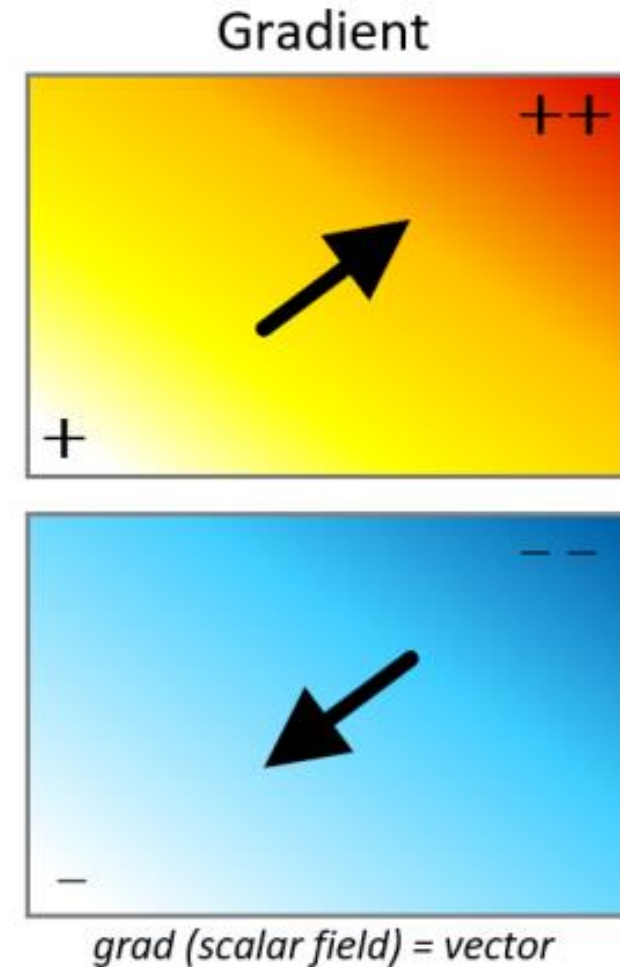
Vector Differential Operator

$$\nabla = \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right)$$

- Think of ∇ as “the direction and rate of change” operator
 - It tells you how things change locally in space (and sometimes time)
- Depending on what you apply it to, it reveals different aspects of variation

Gradient: Apply ∇ to a **multivariate** scalar function $f(x,y,z)$

- You get the direction of steepest ascent and how steep it is
- Example: In a temperature field, the gradient points to where it gets hotter fastest



The Chain Rule

- Computing the derivative of a composite function:

- $f(x) = u(v(x)) \quad \frac{df}{dx} = \frac{du}{dv} \cdot \frac{dv}{dx}$

- $f(x) = u(v(w(x))) \quad \frac{df}{dx} = \frac{du}{dv} \cdot \frac{dv}{dw} \cdot \frac{dw}{dx}$

Multivariate Chain Rule

- $z_1 = g_1(x)$
- $\vdots$
- $z_M = g_M(x)$
- $y = f(z_1, \dots, z_M)$

$$\frac{\partial y}{\partial x} = \sum_{j=1}^M \frac{\partial y}{\partial z_j} \frac{\partial z_j}{\partial x}$$

Calculus Chain Rule

Scalar:

$$z = g(x)$$

$$y = f(z)$$

$$\frac{dy}{dx} = \frac{dy}{dz} \frac{dz}{dx}$$

Multivariate:

$$\mathbf{z} = g(x)$$

$$y = f(\mathbf{z})$$

$$\frac{dy}{dx} = \sum_j \frac{\partial y}{\partial z_j} \frac{\partial z_j}{\partial x}$$

Calculus Chain Rule

• Scalar:

• $z = g(x)$

• $y = f(z)$

• $\frac{dy}{dx} = \frac{dy}{dz} \frac{dz}{dx}$

Multivariate:

$\mathbf{z} = g(x)$

$y = f(\mathbf{z})$

$\frac{dy}{dx} = \sum_j \frac{\partial y}{\partial z_j} \frac{\partial z_j}{\partial x}$

Multivariate:

$\mathbf{z} = g(\mathbf{x})$

$\mathbf{y} = f(\mathbf{z})$

$\frac{dy_i}{dx_k} = \sum_j \frac{\partial y_i}{\partial z_j} \frac{\partial z_j}{\partial x_k}$

Break time

03:00

Review of Probability

Why Uncertainty Matters

- Uncertainty occurs when we have:
 - Partial observability
AND/OR
 - Nondeterministic environments
 - Random events
 - Other agents acting
- A rational agent may need to weigh:
 - Relative importance of various goals
 - Likelihood that (degree to which) they will be achieved
- We will use probability to handle uncertainty in decision making

Basic Vocabulary

- **experiment** – procedure that generates some random result
- **outcome** – possible result of an experiment (a "possible world" in R&N)
 - Lower-case letter or word; ω , ω_1 , etc.
- **sample space** – set of all possible outcomes
 - Ω
- **random variable** – (informally) holds some **outcome**
 - Upper-case letter or word
 -
- **event** – subset of the sample space; set of outcomes
 - Lower-case letter or word

- Rolling a 6-sided die one time.
- "I rolled a 2."
- $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Let X be the outcome of rolling the die.
 $X=2$ is the outcome "I rolled a 2."
- $\{2\}$ is the event "I rolled a 2".
 $\{2, 4, 6\}$ is the event "I rolled an even number."

The Sample Space

- Can be discrete...

- Finite...

$\{1,2,3,4,5,6\}$

- ... or countably infinite

$\mathbb{Z}^+$ (positive integers)

- ... or continuous.

$\mathbb{R}$

$[1, 3]$

We'll focus on *finite* sample spaces.

Probability model

- A *probability model* is a mathematical representation of a random phenomenon.
 - It describes how to calculate the probability of any event.
 - A probability model may involve one or more *probability distributions*.

Probability model

- A **probability distribution** is characterized by:
 1. Its sample space Ω
 2. A function P :
 - If sample space is discrete (finite or countably infinite):
 P is a **probability mass function** (PMF)
that assigns a probability to each **outcome**.
 - If sample space is continuous:
 P is a **probability density function** (PDF)
where the integral of the PDF over an interval gives probability of **outcome** in that interval.
- From the function, we can also compute the probabilities of **events**.

Finite Probability Distribution

outcome: ω , a single outcome, like 2
sample space: Ω , set of all possible outcomes
event: a set of outcomes, like $\{2\}$ or $\{2, 4, 6\}$

- Let e be some **event**. The probability of the event is written $P(e)$.
 - Shorthand: For an **outcome** ω , we often write $P(\omega)$ instead of $P(\{\omega\})$

- A **finite probability distribution** specifies the probability of each possible **outcome** from a finite sample space.

- ... We might instead say "of each possible **event**", or $P(\text{some event})$ can be calculated:

Fair 6-sided die roll

$$P(1) = 1/6$$

$$P(2) = 1/6$$

$$P(3) = 1/6$$

$$P(4) = 1/6$$

$$P(5) = 1/6$$

$$P(6) = 1/6$$


$$P(\{2\}) \text{ or } P(\{2, 4, 6\})$$

P must follow the Axioms of Probability

Also called the Kolmogorov Axioms

outcome: ω , a single outcome, like 2
sample space: Ω , set of all possible outcomes
event: a set of outcomes, like $\{2\}$ or $\{2, 4, 6\}$

1. $\forall \omega P(\omega) \geq 0$

Every outcome has a non-negative probability.

2. $P(\Omega) = 1$

$P(\Omega)$ represents probability of any of the entire set of possible outcomes occurring.

When Ω is finite, this is equivalent to: $\sum_{\omega \in \Omega} P(\omega) = 1$

Together, can conclude:

$$\forall \omega 0 \leq P(\omega) \leq 1$$

Dice roll:

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$P(\Omega) = P(\{1, 2, 3, 4, 5, 6\})$$

$$= P(1) + P(2) + P(3) + P(4) + P(5) + P(6)$$

$$= 1/6 + 1/6 + 1/6 + 1/6 + 1/6 + 1/6$$

$$= 1$$

3. ... to be discussed soon. Let's establish some ideas first.

Logical Operators on Events

- Let a be the event "Tomorrow, it will rain."
Let b be the event "Tomorrow, it will be cold."
- Each event is (will be) either true or false.
It's uncertain, because we don't know the weather tomorrow.
- So we can discuss $P(a)$ as the probability that it will rain tomorrow.
- We can also consider events such as ...
 - $a \wedge b$ "Tomorrow, it will rain and will be cold."
 - $a \vee b$... or
 - $\neg a$... not
 - ...
- ... and probabilities like $P(a \wedge b)$,
the probability that tomorrow it rains and is cold...

Logical View

Events

Set-Theoretic View

x: "I rolled a 4." {4}
 y: "I rolled a 1, 2, or 3." {1, 2, 3}
 z: "I rolled a 2, 3, or 4." {2, 3, 4}

x $\vee$ y
 "I rolled a 4, **or** I rolled a 1, 2, or 3."
 "I rolled a 1, 2, 3, or 4."
 Probability: $4/6 = 2/3$

x $\cup$ y
 $\{4\} \cup \{1, 2, 3\} = \{1, 2, 3, 4\}$

x $\wedge$ z
 "I rolled a 4, **and** I rolled a 2, 3, or 4."
 "I rolled a 4."
 Probability: $1/6$

x $\cap$ z
 $\{4\} \cap \{2, 3, 4\} = \{4\}$

x $\wedge$ y
 "I rolled a 4, **and** I rolled a 1, 2, or 3."
impossible...
 Probability: 0

x $\cap$ y
 $\{4\} \cap \{1, 2, 3\} = \emptyset$

x $\vee$ z
 "I rolled a 4, **or** I rolled a 2, 3, or 4."
 "I rolled a 2, 3, or 4."
 Probability: $3/6 = 1/2$

x $\cup$ z
 $\{4\} \cup \{2, 3, 4\} = \{2, 3, 4\}$

Logical View

Events

x: "I rolled a 4." {4}
y: "I rolled a 1, 2, or 3." {1, 2, 3}
z: "I rolled a 2, 3, or 4." {2, 3, 4}

x $\vee$ y "I rolled a 4, **or** I rolled a 1, 2, or 3."
"I rolled a 1, 2, 3, or 4."
Probability: $4/6 = 2/3$

x $\wedge$ z "I rolled a 4, **and** I rolled a 2, 3, or 4."
"I rolled a 4."
Probability: $1/6$

x $\wedge$ y "I rolled a 4, **and** I rolled a 1, 2, or 3."
impossible...
Probability: 0

x $\vee$ z "I rolled a 4, **or** I rolled a 2, 3, or 4."
"I rolled a 2, 3, or 4."
Probability: $3/6 = 1/2$

Thinking in terms of sets
can be helpful,

but we shouldn't mix the two notations
in what we *write*.

R&N use the logical notation,
so we'll stick with that.

$P(x \wedge y)$ is often written $P(x, y)$

Axioms of Probability

- Important to remember!

1. $0 \leq P(a) \leq 1$

2. $P(\Omega) = 1$

3. $P(a \cup b) = P(a) + P(b)$

Given mutually exclusive events a and b

Example (using prob. of prev. slide)

$$P(\{2, 4, 6\}) = P(2) + P(4) + P(6)$$

Then, $P(\neg a) = 1 - P(a)$

This can be proved from axioms 1-3! (good practice at home 😊)

outcome: ω , a single outcome, like 2
sample space: Ω , set of all possible outcomes
event: a set of outcomes, like $\{2\}$ or $\{2, 4, 6\}$

Random Variables

Informally, a **random variable** is a variable that holds some **outcome** from the sample space.

Experiment: Rolling a 6-sided die one time.
Let R be the outcome of rolling the die.

$R=2$
is the **outcome**
"I rolled a 2."

$P(R=2)$ is the probability of rolling a 2.
May be abbreviated $P(2)$ when the random variable is clear from the context.

$R=2 \vee R=4 \vee R=6$
is the **event**
"I rolled an even number."

$P(R=2 \vee R=4 \vee R=6)$ is the probability of rolling an even number.
We might abbreviate with something like $P(\text{even})$ when clear...

In general, for random variable X we may speak of $P(X = x)$, or $P(X = x_1)$, $P(X = x_2)$, etc. (Or abbreviate $P(x_1)$, $P(x_2)$, etc., when context is clear.)

Random Variables

Notation often varies for
Boolean random variables.

outcome: ω , a single outcome, like 2
sample space: Ω , set of all possible outcomes
event: a set of outcomes, like $\{2\}$ or $\{2, 4, 6\}$

Experiment: Is it raining?

Let Rain be a **Boolean** random variable – either it's raining, or it isn't.

Rain=true
is the **outcome**
"It's raining."

$P(\text{Rain}=\text{true})$ is the probability that it is raining.
May be abbreviated $P(\text{rain})$.

Lower case rain means Rain=true.
Upper case Rain means the random variable.

Rain=false
is the **outcome**
"It's not raining. "

$P(\text{Rain}=\text{false})$ is the probability that it is **not** raining.
May be abbreviated $P(\sim\text{rain})$ or $P(\neg\text{rain})$

Expected Value (Expectation)

- The average value of a random variable is called the *expected value* or *expectation*. For a discrete random variable, it is:

$$E[X] = \sum_x P(X = x)x$$

Sometimes the symbol μ or μ_X is used instead of $E[X]$. Note that this is a weighted average, weighted by probability.

- The following properties hold for expected value, for constant α and random variables X and Y :

$$E[\alpha] = \alpha$$

$$E[\alpha X] = \alpha E[X]$$

$$E[\alpha + X] = \alpha + E[X]$$

$$E[X + Y] = E[X] + E[Y]$$

$$E[X] = \sum_x P(X = x)x$$

Variance

- The *variance* of a random variable is a measure of how much it varies – how spread out it is – around its average value. We can write it as $Var(X)$ or σ^2 :

$$\begin{aligned}\sigma^2 &= E[(X - E[X])^2] \\ &= \sum_x (x - E[X])^2 P(X = x)\end{aligned}$$

Mutual Exclusivity

- Two events are *mutually exclusive* iff they can't both occur.

x: "I rolled a 4." {4}

y: "I rolled a 1, 2, or 3." {1, 2, 3}

x $\wedge$ y: "I rolled a 4, **and** I rolled a 1, 2, or 3."
impossible...

The event x and the event y are mutually exclusive:
 $P(x \wedge y) = 0$.

x $\cap$ y: "I rolled a 4, **and** I rolled a 1, 2, or 3."
impossible...

$\{4\} \cap \{1, 2, 3\} = \emptyset$

There are no outcomes in this event!
It's impossible!

Two events a and b are mutually exclusive iff $P(a \wedge b) = 0$.

Axioms of Probability

outcome: ω , a single outcome, like 2
sample space: Ω , set of all possible outcomes
event: a set of outcomes, like $\{2\}$ or $\{2, 4, 6\}$

The probability measure P of a probability model must follow the **axioms of probability**:

Also called the
Kolmogorov Axioms

1. $\forall \omega P(\omega) \geq 0$ Every outcome has a non-negative probability.
2. $P(\Omega) = 1$ $P(\Omega)$ represents probability of any of the entire set of possible outcomes occurring.
When Ω is finite, this is equivalent to: $\sum_{\omega \in \Omega} P(\omega) = 1$

Together, can conclude:
 $\forall \omega 0 \leq P(\omega) \leq 1$

3. For every finite collection of mutually exclusive events $a_1, a_2, \dots, a_k$:
 $P(a_1 \vee a_2 \vee \dots \vee a_k) = P(a_1) + P(a_2) + \dots + P(a_k)$

Equivalently:

$$P\left(\bigvee_{i=1}^k a_i\right) = \sum_{i=1}^k P(a_i)$$

When $k=2$, we have:
 $P(a \vee b) = P(a) + P(b)$
again assuming a and b are mutually exclusive.

outcome: ω , a single outcome, like 2
sample space: Ω , set of all possible outcomes
event: a set of outcomes, like $\{2\}$ or $\{2, 4, 6\}$

Events and Outcomes

- Axiom 3:

For every finite collection of mutually exclusive events $b_1, b_2, \dots, b_k \in \Omega$:

$$P(b_1 \vee b_2 \vee \dots \vee b_k) = \sum_{i=1}^k P(b_i)$$

- If we consider just a single event $a = \{\omega_1, \omega_2, \dots, \omega_k\}$, we can also think of it as a disjunction of outcomes:

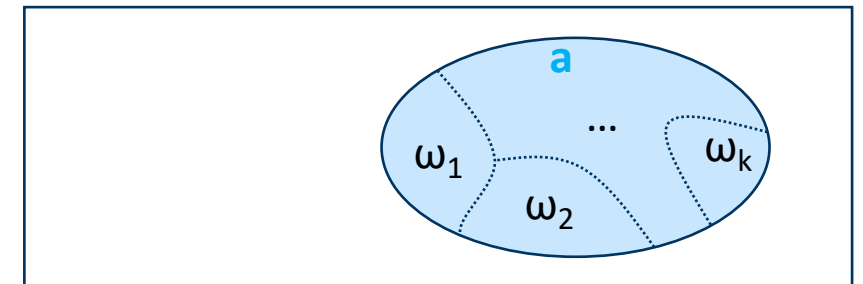
$$a = \omega_1 \vee \omega_2 \vee \dots \vee \omega_k \quad \Omega$$

- Outcomes are inherently mutually exclusive, so:

$$P(a) = P(\omega_1 \vee \omega_2 \vee \dots \vee \omega_k) = \sum_{i=1}^k P(\omega_i)$$

Equivalently:

$$P(a) = \sum_{\omega \in a} P(\omega)$$

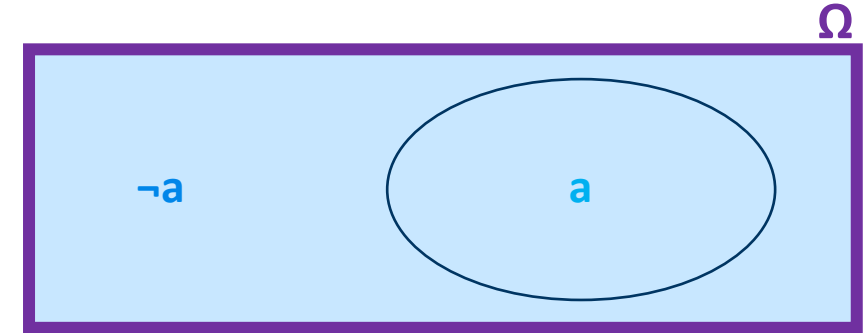


Example: Proof from Axioms of Probability

1. $\forall \omega P(\omega) \geq 0$
2. $P(\Omega) = 1$. For a finite set Ω , this is equivalent to $\sum_{\omega \in \Omega} P(\omega) = 1$.
3. For every finite collection of mutually exclusive events $a_1, a_2, \dots, a_k$, we have $P(a_1 \vee a_2 \vee \dots \vee a_k) = P(a_1) + P(a_2) + \dots + P(a_k)$.
Equivalently, $P(\bigvee_{i=1}^k a_i) = \sum_{i=1}^k P(a_i)$.

- Claim: $P(\neg a) = 1 - P(a)$
- Proof:

$$\begin{aligned}
 P(\neg a) &= \sum_{\omega \in \neg a} P(\omega) \\
 &= \sum_{\omega \in \neg a} P(\omega) + \overbrace{\sum_{\omega \in a} P(\omega) - \sum_{\omega \in a} P(\omega)}^0 \\
 &= \sum_{\omega \in \Omega} P(\omega) - \sum_{\omega \in a} P(\omega) \\
 &= 1 - \sum_{\omega \in a} P(\omega) \\
 &= 1 - P(a)
 \end{aligned}$$



Just adding 0

By axiom 3, since Ω is the set of all outcomes.

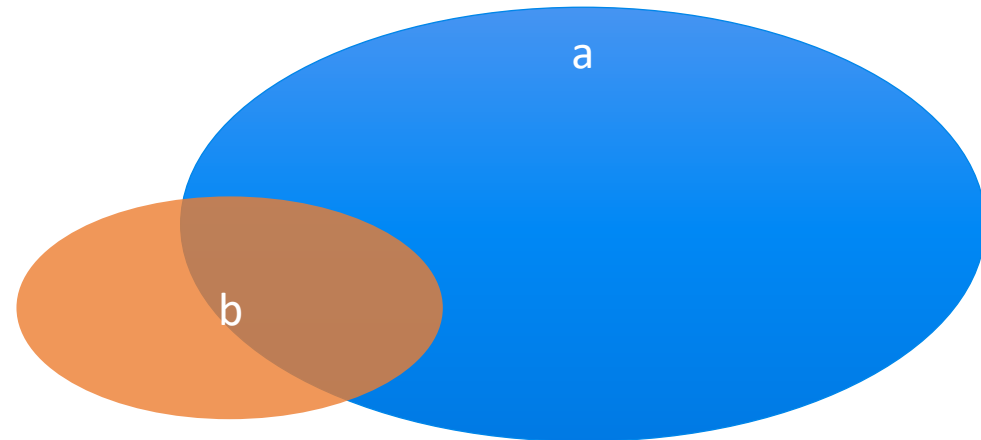
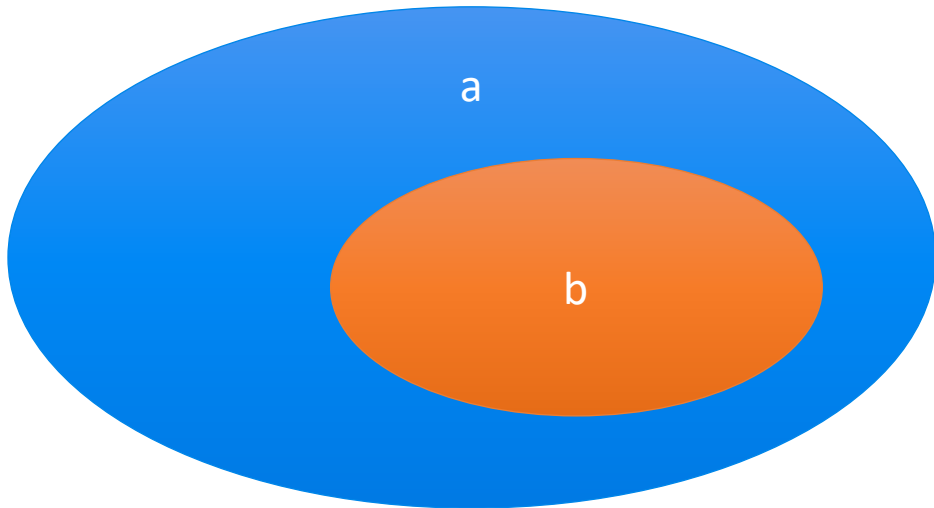
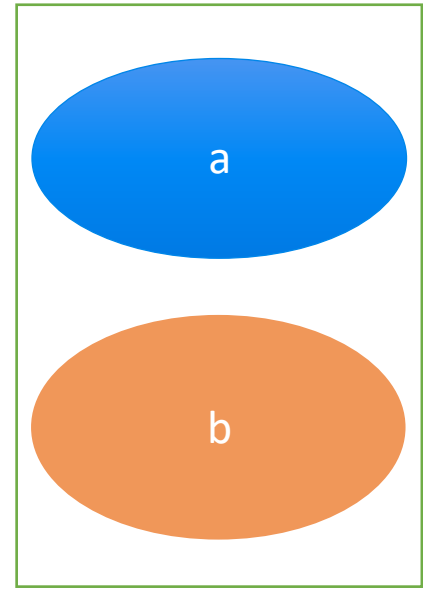
By axiom 2 (finite set Ω)

By axiom 3

Inclusion-Exclusion Principle

- If events are mutually exclusive: $P(a \vee b) = P(a) + P(b)$
- What if the events aren't mutually exclusive?

$$P(a \vee b) = P(a) + P(b) - P(a \wedge b)$$



Types of Probabilities

- $P(X = x)$ – unconditional or **prior probabilities**
 - Refer to degrees of belief in the absence of other information!
 - Usually we have some **evidence**
- Conditional probabilities
 - Probability given that we know something has happened
 - **Example: $P(a | b)$**
 - Doesn't invalidate the prior information
 - Note: we will come back to them later!

More on Conditional Probability

- $P(X = x \mid Y = y)$, or just $P(x \mid y)$

conditional probability or *posterior probability*

- Probability that x happened given (or assuming) that y happened.

$$P(x \mid y) = \frac{P(x \wedge y)}{P(y)}$$

Only defined if $P(y) > 0$.
If $P(y) = 0$, then $P(x \mid y)$ doesn't make sense,
since we can't "assume" an impossible thing (y) happened.

$$P(x \mid y) = \frac{P(x, y)}{P(y)}$$

- Often rewritten as the *product rule*:

$$P(x \wedge y) = P(y) P(x \mid y)$$

$$P(x, y) = P(y) P(x \mid y)$$

Chain Rule (General Product Rule)

Product rule: $\mathbf{P}(X, Y) = \mathbf{P}(X) \mathbf{P}(Y | X)$

Similarly: $\mathbf{P}(A, B, C, D) = \mathbf{P}(A, B, C) \mathbf{P}(D | A, B, C)$


$$= \mathbf{P}(A, B) \mathbf{P}(C | A, B) \mathbf{P}(D | A, B, C)$$

$$= \mathbf{P}(A) \mathbf{P}(B | A) \mathbf{P}(C | A, B) \mathbf{P}(D | A, B, C)$$

You can do any ordering you want, with any number of variables.

$$\mathbf{P}(X_1, X_2, \dots, X_n) = \mathbf{P}(X_1) \mathbf{P}(X_2 | X_1) \mathbf{P}(X_3 | X_1, X_2) \dots \mathbf{P}(X_n | X_1, \dots, X_{n-1})$$

$$= \prod_{i=1}^n \mathbf{P}(X_i | X_1, \dots, X_{i-1})$$

Bayes' Rule (aka Bayes' Theorem)

We already know:

$$\begin{aligned}\mathbf{P}(A, B) &= \mathbf{P}(A \mid B) \mathbf{P}(B) \\ &= \mathbf{P}(B \mid A) \mathbf{P}(A)\end{aligned}$$

Equating the right-hand sides and rearranging, we get *Bayes' Rule*:

Written with probability distributions:

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(B \mid A) \mathbf{P}(A)}{\mathbf{P}(B)}$$

Written with probabilities of particular outcomes:

$$\begin{aligned}P(A = a \mid B = b) &= \frac{P(B = b \mid A = a)P(A = a)}{P(B = b)} \\ P(a \mid b) &= \frac{P(b \mid a)P(a)}{P(b)}\end{aligned}$$